Alert-Budgeted Clinical Early Warning: Conformal Risk Control for Stateful Deterioration Alarm Policies

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Abstract

Clinical early warning requires an explicit balance between detecting deterioration and generating a manageable alert load. We propose to express that balance as a budget on the patient-average number of alarms per day and to calibrate a causal alarm policy with refractory and escalation state. The reference procedure aggregates dependent alarm sequences into patient-level losses and applies simultaneous finite-grid upper-confidence bounds under independent, identically distributed patient sampling, with the scorer, policy family and support bound fixed before calibration. A candidate is deployed only when its raw empirical loss plus a Hoeffding or empirical-Bernstein radius meets the budget; otherwise an exact no-alarm policy provides a safe fallback. The artificial-intelligence contribution is policy-level risk calibration that accounts for internal state, while the engineering contribution is an auditable operating point specified by an alert budget rather than an arbitrary threshold. The supplied implementation includes causal reference learners, official-format intensive-care data readers, and a synthetic effective-sample-size diagnostic. The proposed clinical protocol distinguishes population risk from finite-cohort exceedance, reports sensitivity alongside conditional detection lead time, and treats cross-hospital transfer without target calibration as distribution shift rather than guaranteed deployment. Real-cohort comparative performance and clinical effectiveness remain to be established.

Keywords: Clinical early warning, Alarm budgets, Stateful policies, Risk calibration, Clinical time series

1. Introduction

Deterioration events in hospitalised patients unfold over hours, and the physiological trajectory that precedes them is recorded long before the event

itself is recognised. This temporal structure has motivated a substantial literature on data-driven early warning, ranging from static severity scores to sequential models that forecast deterioration trajectories from electronic health records [1] and compact, efficiency-aware networks that predict sepsis onset from structured time series [2]. Reported discrimination is high and improving, with in-hospital mortality and onset prediction now routinely exceeding the performance of classical scoring instruments [3]. The implicit promise of this line of work is that a sufficiently accurate model, once installed at the bedside, will shorten time to intervention.

Discriminative prediction and operational alerting address different questions. A score-to-alert conversion requires a decision rule, whereas threshold-independent ranking metrics do not specify how frequently that rule emits alarms. Reviews of intensive-care adverse-event prediction discuss incomplete reporting of alert burden, time-to-alert and deployment calibration [4]. Reviews of multivariate clinical time-series learning also motivate evaluation beyond architectural comparisons [5, 6]. These concerns support reporting the alarm policy and its realised load explicitly, rather than treating an area-under-the-curve comparison as a specification of bedside alerting behaviour.

Recent research improves the backbone that produces the risk score. For irregularly sampled and heterogeneously missing observations, approaches include reliability-aware temporal aggregation [7, 8], graph formulations over irregular observations [9], and masked-attention models for deterioration prediction [10]. Transfer-oriented variants align feature spaces across cohorts [11], while dynamic temporal graphs incorporate clinical calculators into the prediction pathway [12]. These methods motivate prospective backbone comparisons, but their adapters are not included in the current reference implementation. Our implemented contribution instead concerns the causal mapping from fixed scores to stateful alarms and the risk associated with selecting that mapping.

A separate literature attaches reliability estimates to clinical predictions. Selective prediction and calibration have been analysed for multimodal clinical classification [13], reliability estimation has been extended beyond output-space calibration for time-series classifiers [14], and conformal methods have been used for physiological regression uncertainty [15]. Our distinction is the unit and estimand of alarm-policy calibration: refractory state makes successive emitted alarms dependent, so those hours cannot simply be counted as independent patient samples. This does not imply that every conformal treatment of time series is invalid or that a stateful policy cannot be cali-

brated at the trajectory level. The reference construction uses independent patient blocks and fixed finite-grid upper-confidence bounds, a stronger sampling assumption than exchangeability alone.

This draft makes the alarm budget an explicit constraint and calibrates the policy that generates alerts. The implementation separates causal score estimation from full-trajectory policy replay. Its finite-sample bound concerns the population mean of patient-level rates, not a hard maximum for each patient or ward. Real-cohort comparisons remain prospective, and the current contributions are as follows.

- We formulate early warning as a stateful alarm policy in which the refractory window and escalation tier are control variables, so that alarm volume becomes a design parameter of the monitoring system rather than an emergent property of a threshold.
- We derive a patient-blocked finite-grid upper-confidence bound on the population alarm rate under independent sampling, and distinguish it from a deliberately invalid timestep effective-sample-size shortcut.
- We provide causal reference learners and official-format data readers, and specify a prospective evaluation protocol under shared requested budgets, with cross-hospital transfer and observation sparsification as planned extensions.

The remainder of the paper reviews related work (Section 2), develops the policy formulation and the calibration procedure (Section 3), and specifies the datasets, baselines, metrics, and experimental protocol (Section 4).

2. Related Work

2.1. Backbones for Irregularly Sampled Clinical Time Series

Recent work models irregular sampling explicitly, using continuous-time formulations [8], reliability-aware temporal aggregation [7], and graphs over observations and variables [9]. Complementary approaches model the observation process to synthesise irregular trajectories [16] or exploit informative missingness for ward deterioration prediction [10]. Domain-specific applications include venous thromboembolism detection after cancer treatment [17], while surveys synthesise the wider modelling landscape [18]. These methods

are potential future score backbones, not models already integrated or reproduced in the current package. Any integration must retain prefix-only information and cannot reuse a full-trajectory imputation procedure without a separate causality audit.

2.2. Early Warning and Deterioration Prediction

Clinical early warning has moved from static severity scores toward dynamic, trajectory-aware prediction. Recent multicentre work predicts the shape of sepsis deterioration rather than a single risk estimate, arguing that heterogeneous trajectories require patient-specific forecasting [1], while efficiency-oriented designs pursue compact models that remain explainable under the compute constraints of a ward workstation [2]. Others inject clinical reasoning into the prediction pathway: dynamic temporal graphs embed calculator-derived features so that the model inherits the semantics of established instruments [12], and language-model-guided simulation is used to generate interpretable physiological dynamics for early warning [19]. Transfer-oriented formulations address the practical reality that a model trained in one institution must be reused in another with different coding practice and case mix [11], and earlier work established that deep models over longitudinal inpatient records can outperform tabular baselines for outcome prediction [3, 20, 21]. Across this literature the model is the unit of contribution; the alerting rule that converts its output into an operational signal is treated as an afterthought, and comparison between methods is almost always conducted at a fixed threshold rather than at a fixed alarm volume.

2.3. Calibration, Selective Prediction, and Conformal Guarantees

Calibration and selective prediction have been studied in multimodal clinical classification [13]. Reliability estimation for time-series classifiers also examines evidence beyond output remapping [14], while adaptation audits consider realistic label delays [22]. Conformal methods have been applied to physiological regression uncertainty [15], and inference-time adaptation has addressed patient-specific shift in haemodynamic prediction [23]. These contributions concern different units, targets and sampling assumptions; their guarantees cannot be declared invalid merely because an unrelated downstream alarm rule has memory. The present reference procedure instead specifies a patient-average alarm-rate estimand and uses independent-block concentration, without claiming the weaker assumptions or different risk statement of standard conformal constructions.

2.4. Evaluation Practice

Risk-of-bias instruments for prediction models address artificial-intelligence-specific issues in calibration, heterogeneity and applicability [24]. Reviews of intensive-care adverse-event and time-series prediction also discuss reporting and evaluation limitations [4–6]. These concerns motivate a proposed protocol that reports requested budgets, realised alarm rates, sensitivity and lead time alongside discrimination. The real-cohort comparison is still prospective, so its design should not be mistaken for evidence that every planned measurement has already been produced. Shared requested budgets make the operational constraint explicit but do not imply identical realised alert volume across models.

3. Method

3.1. Problem Setting and Notation

Let patient i contribute observations $\mathcal{O}_i = \{(t_{ij}, m_{ij}, v_{ij})\}_{j=1}^{n_i}$, with observation times t_{ij}, variable indices m_{ij}, and values v_{ij}. The number and spacing of observations may differ across patients and variables. Decisions are made on a prespecified regular grid with sampling interval Δ hours, using only observations available at each decision time. A causal score function maps this history to $s_{it} = f_\theta(\mathcal{O}_{i,\le t}) \in [0, 1]$; any preprocessing, including imputation, must satisfy the same restriction.

An observed event has time $t_i^\star$, and a useful early alarm falls in $[t_i^\star - W, t_i^\star)$, where $W = 12$ hours is a design choice for the proposed evaluation. Records without an observed event are represented separately from records whose event time is outside observed follow-up. Clinical event time and the boundary of a shifted prediction label are not interchangeable. The data protocol in Section 4 therefore requires an explicit onset convention and does not evaluate lead time against an unobserved or left-censored clinical onset.

3.2. Stateful Alarm Policies

The policy $\pi_\lambda = (\lambda, \tau, \kappa)$ comprises a threshold, an integer refractory duration, and an integer escalation duration. Here $\tau \ge 0$ is the number of subsequent decision steps suppressed after an alarm; $\kappa = 0$ disables escalation, while $\kappa > 0$ permits an override after κ consecutive above-threshold steps following the last alarm. Let z_{it} be the number of refractory steps remaining at the start of step t, and let c_{it} include the current exceedance

and the consecutive suppressed exceedances immediately preceding it. With initial state zero, the emission and update are

$$a_{it} = \mathbf{1}[s_{it} \text{ finite},\ s_{it} \geq \lambda,\ z_{it} = 0\ \vee\ (\kappa > 0\ \wedge\ c_{it} \geq \kappa)]\,,$$
$$z_{i,t+1} = \begin{cases} \tau, & a_{it} = 1, \\ \max(0, z_{it} - 1), & a_{it} = 0. \end{cases} \tag{1}$$

The exceedance counter resets after an alarm or a non-exceedance, including a missing score. The refractory duration counts decision steps, not observations in the irregular raw record. Setting $\tau = 0$ gives the memoryless threshold policy, whereas $\lambda = +\infty$ gives an exact no-alarm policy. The latter is an explicit safe fallback and is distinct from the largest finite grid threshold, which can still emit alarms.

The scorer and structural parameters are selected using training and validation patients and are then fixed before risk calibration. Every candidate threshold is evaluated by replaying the entire causal policy, with state reset at each patient's admission. Within-patient alarm indicators are dependent, so counting them as independent calibration units is not justified. Dependence within a patient does not itself prevent calibration over independently sampled patient trajectories.

3.3. Risk Estimand and the Timestep Shortcut

For n_i decision steps, define exposure $D_i = n_i\Delta/24$ days and alarm count $A_i(\lambda) = \sum_t a_{it}(\lambda)$. The target is the patient-average daily alarm rate

$$R(\lambda) = \mathbb{E}[A_i(\lambda)/D_i]\,. \tag{2}$$

This estimand weights each patient equally and is not the pooled ward rate $\mathbb{E}[A_i]/\mathbb{E}[D_i]$, which weights patients by exposure. Both may be clinically relevant, but their identification requires different loss definitions. The present certificate concerns only Equation (2); it does not guarantee a fixed maximum count for every patient or a fixed realised total for a ward.

The synthetic comparison includes a deliberately invalid effective-sample-size shortcut. It uses the same empirical patient-average losses as the blocked selector, but substitutes the number of calibration timesteps for the number of calibration patients in the confidence radius. This isolates the consequence of inflating the unit of information without changing the point estimator. It is not an implementation of a particular published timestep-conformal algorithm, nor does the comparison establish that every alternative time-series calibration method is invalid.

3.4. Patient-Blocked Uniform Risk Bounds

Define the patient loss as the observed daily alarm rate,

$$L_i(\lambda) = A_i(\lambda)/D_i \in [0, L_{\max}]. \tag{3}$$

The upper bound must hold over the deployment population's support, not merely over the observed calibration sample. If eligible records have at least $H_{\min}$ hours of exposure, let $n_{\min} = \lceil H_{\min}/\Delta \rceil$ and $g = \tau + 1$ without escalation, or $g = \min(\tau + 1, \kappa)$ when escalation is enabled. The minimum firing separation is g, so the exact discrete-support upper bound is

$$L_{\max} = \sup_{n \in \mathbb{N},\ n \geq n_{\min}} \frac{24}{n\Delta} \left(1 + \left\lfloor \frac{n-1}{g} \right\rfloor \right). \tag{4}$$

The rate decreases between firing-count increments, and its peaks at $n = 1 + qg$ decrease as q increases; hence checking $n_{\min}$ and the next such peak is sufficient. For hourly sampling, $\tau = 4$, no escalation and $H_{\min} = 24$, a 26-hour stay yields $L_{\max} = 144/26$, exceeding the five alarms per day possible over exactly 24 hours. Escalation can increase this bound and must not be omitted from its calculation. Losses outside the declared interval are rejected, not clipped, because clipping would certify a different estimand.

Condition on independently obtained training and validation data. The scorer, structural parameters, finite threshold grid Λ, and support bound are fixed under this conditioning. Calibration trajectories are independent and identically distributed draws from the same population as future deployment trajectories, although each trajectory may have arbitrary internal dependence. Exchangeability alone is not sufficient for the independent-sample concentration argument used here.

Proposition 1 (Budget validity)**.** *Let* $\widehat{R}_n(\lambda) = n^{-1} \sum_{i=1}^{n} L_i(\lambda)$ *and* $\alpha \in (0, 1)$*. Under the conditions above, select*

$$\widehat{\lambda} = \min \left\{ \lambda \in \Lambda : \widehat{R}_n(\lambda) + L_{\max} \sqrt{\frac{\log(|\Lambda|/\alpha)}{2n}} \leq B \right\} \tag{5}$$

when the set is nonempty, and select the exact no-alarm policy $\widehat{\lambda} = +\infty$ *otherwise. For a finite nonnegative budget* B*, the selected policy satisfies*

$$\Pr\{R(\widehat{\lambda}) > B\} \leq \alpha. \tag{6}$$

The probability is over calibration samples; R *is a population expectation conditional on the fixed scorer and policy family.*

Proof. For each fixed λ, the relevant one-sided Hoeffding inequality is

$$\Pr\{R(\lambda) - \widehat{R}_n(\lambda) > \epsilon\} \leq \exp(-2n\epsilon^2/L_{\max}^2).$$

Setting $\epsilon = L_{\max}\sqrt{\log(|\Lambda|/\alpha)/(2n)}$ and applying a union bound gives simultaneous upper bounds $R(\lambda) \leq \widehat{R}_n(\lambda) + \epsilon$ for all grid policies with probability at least $1 - \alpha$. On that event, any policy selected from the feasible set satisfies $R(\widehat{\lambda}) \leq B$. If the feasible set is empty, the fallback has identically zero loss and hence satisfies the budget without a concentration argument. Thus a budget violation can occur only on the failure event of the simultaneous upper bounds, which has probability at most α. □

For $n \geq 2$, the implementation also permits the empirical-Bernstein radius[1]

$$\epsilon_{\mathrm{EB}}(\lambda) = \widehat{\sigma}(\lambda)\sqrt{\frac{2\log(2|\Lambda|/\alpha)}{n}} + \frac{7L_{\max}\log(2|\Lambda|/\alpha)}{3(n-1)}, \tag{7}$$

where $\widehat{\sigma}$ is the sample standard deviation with denominator $n - 1$. This bound can improve on Hoeffding when variance is small and the sample is sufficiently large, but it is not uniformly tighter. Both variants are finite-grid upper-confidence risk procedures; patient blocking does not turn either into the standard expected-risk conformal construction. The manuscript's conformal terminology denotes this broader calibration motivation, not a claim that the proof requires only conformal exchangeability.

The certificate always uses the raw empirical loss at the deployed threshold. Stateful alarm counts are not assumed to be monotone in the threshold, so isotonic regression is optional descriptive output and never substitutes for that loss in the upper bound. A finite test-cohort mean can exceed the budget even when its population expectation does not, so its exceedance frequency is an empirical diagnostic rather than the exact probability in Equation (6). Likewise, calibration transferred across distributions is not covered without the same-population assumption.

3.5. Operating Points and Lead-Time Evaluation

The reference selector deploys the smallest finite threshold whose raw risk upper bound meets the requested budget. This is a deterministic operating-

[1]The finite-class empirical-Bernstein bound and its constants are given by Maurer and Pontil, *Empirical Bernstein Bounds and Sample Variance Penalization*, https://arxiv.org/abs/0907.3740.

point convention rather than a guarantee of optimal detection. A separate utility objective could select among the uniformly certified candidates without invalidating the risk bound, but such an optimiser is not implemented here. Neither a lower threshold nor a higher budget is assumed to improve every operational metric of a stateful policy.

For an event observed during follow-up, define the detection window $\mathcal{W}_i = [\max(0, t_i^\star - W), t_i^\star)$. An event is detected if at least one alarm falls in that window, and its lead time is

$$\ell_i(\lambda) = t_i^\star - \min\{t \in \mathcal{W}_i : a_{it}(\lambda) = 1\}. \tag{8}$$

Lead time is undefined for undetected events and is represented as missing rather than zero. The reported median therefore describes detected events only and must always be accompanied by sensitivity and the number of eligible events. Events at admission, outside observed follow-up, or without an identifiable onset cannot supply an observed pre-onset lead time. A retrospective top-k selection over a complete trajectory is noncausal and may serve only as a labelled oracle comparator, not as a deployable early-warning policy.

Budget sweeps compare methods under the same requested upper-risk constraint, not necessarily at identical realised alarm rates. The plotted frontier uses realised alarm rate and conditional median lead time, retaining only nondominated pairs. At equal alarm rates only the largest lead time is retained; at larger rates a point remains only when its lead time strictly improves. Because conditional lead time can improve by missing difficult cases, sensitivity must be reported alongside the frontier before any clinical interpretation.

4. Experimental Setup

4.1. Status and Data Sources

This section specifies the proposed real-cohort evaluation rather than completed clinical results. The supplied implementation supports data ingestion, two reference score learners, synthetic diagnostics, and cached-score budget evaluation. External backbone adapters, the full comparative grid, and clinical effectiveness experiments remain incomplete. Consequently, no real-cohort performance or measured accelerator-hour budget is claimed in this draft.

Algorithm 1 Patient-blocked upper-confidence calibration of an alarm policy

Require: independent calibration patients; fixed scorer, grid Λ, structural parameters, valid $L_{\max}$; budget B; level α

for $\lambda \in \Lambda$ **do**
    replay the causal policy on each patient's observed follow-up
    compute $L_i(\lambda)$ and reject any loss outside $[0, L_{\max}]$
    compute raw $\widehat{R}_n(\lambda)$ and Hoeffding or empirical-Bernstein radius
end for
$\mathcal{F} \leftarrow \{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon(\lambda) \leq B\}$
$\widehat{\lambda} \leftarrow \min \mathcal{F}$ if nonempty; otherwise $+\infty$
return threshold, raw certificate, and explicit fallback status

The public Challenge 2019 release contains 20,336 records in training set A and 20,000 in training set B, from two hospital systems.[2] A third hospital was used as hidden challenge evaluation data and is not a public training subset for an A→C experiment. The intended cross-hospital comparison therefore uses A→B, with the reverse direction as an optional replication. Record identifiers provide record-disjoint partitions but cannot establish whether repeated admissions belong to the same individual, leaving independent patient sampling as an assumption requiring separate justification.

Challenge 2012 supplies an in-hospital mortality task using measurements from the first 48 hours of an intensive-care stay.[3] Its patient files are long-form observations, and mortality outcomes are joined from separate files by RecordID. Measurements are assigned to the next hourly decision at which they are available, with the last valid measurement retained within each interval and no backward filling. The resulting 49 decision points at hours 0 through 48 are a mortality input, not a deterioration follow-up with an invented event onset; cross-task transfer also requires explicit unit conversion and a shared feature schema before any pretrained model can be reused.

[2] Official data and label documentation: https://physionet.org/content/challenge-2019/1.0.0/.

[3] Official format, variable units and separate outcome files: https://physionet.org/content/challenge-2012/1.0.0/.

4.2. Partitioning, Labels and Eligibility

The intended source split is 60% training, 15% validation, 10% calibration and 15% test, with no record crossing partitions. Training and validation determine the scorer, preprocessing, structural parameters and threshold grid before risk calibration. Calibration determines the deployed threshold without test-set tuning, while exposure-based eligibility must be declared independently of the calibration and test samples. An outcome-dependent purge is not needed for record separation and is reserved for explicitly reported sensitivity analyses because it changes the population and may introduce selection bias.

The 2019 SepsisLabel becomes positive six hours before the clinical onset defined by the challenge, so the default reader's label-boundary time must not be called clinical onset. An explicit six-hour shift reconstructs the candidate clinical time only when the label boundary is observed within the record. Initially positive records have an uncertain, left-censored onset and are not assigned a fabricated six-hour clinical time; shifted events outside observed follow-up are also ineligible for observed lead-time evaluation. Padding and post-onset times are excluded from event-label training rather than labelled negative, while alarm-rate evaluation still counts the entire declared observed exposure.

For local target calibration, the scorer and structural policy parameters would remain fixed from the source, with disjoint target calibration and test records. The proposed target sample sizes are $m \in \{50, 100, 200, 500\}$, but feasibility must be reported because small samples may yield only the no-alarm fallback. A source-only transferred threshold is a separate distribution-shift diagnostic and is not assigned the same-population certificate. All comparisons use shared requested budgets, and realised alarm rates are reported rather than assumed equal.

4.3. Score Models and Comparators

The implemented learners are a prefix-feature logistic model, optionally replaced by LightGBM, and a unidirectional masked gated recurrent network. Prefix features use only current or past observations, and future-perturbation tests check causal predictions. The recurrent reference consumes values, observation masks and finite elapsed-time features, with missing static values zero-imputed; it is not a GRU-D reimplementation and currently trains on CPU. Additional architectures are prospective extensions and must pass

the same causality checks before inclusion, particularly when an imputation method ordinarily uses an entire trajectory.

Clinical-score functions in the package are research-only partial instruments, not validated reconstructions of the complete scales. In particular, the available NEWS components omit consciousness and supplemental oxygen, SIRS omits the immature-neutrophil alternative, and SOFA lacks several required components. Their raw scores are not calibrated probabilities and cannot be passed directly to probability-calibration metrics without a training-only probability adapter. Retrospective top-k selection and test-rate-tuned thresholds, if added, must be labelled nondeployable oracle comparators rather than fair prospective baselines.

4.4. Metrics and Uncertainty

The proposed evaluation reports discrimination, calibration, patient-average alarms per day, sensitivity, and lead time among detected eligible events. A lead-time median is accompanied by the detection denominator because improving a conditional median can reflect missing harder cases rather than detecting earlier. Budget diagnostics distinguish the population expectation in Equation (6), the held-out finite-cohort mean, and the per-patient exceedance tail. The synthetic driver repeats partitions of a reused cohort, so its Wilson intervals are descriptive and do not establish independent-binomial coverage or prove the population-risk theorem.

Patient-block bootstrap resampling is the intended uncertainty procedure for correlated hourly predictions. DeLong tests are appropriate only when their observational units satisfy the required independence assumptions, not when every correlated hour is treated as an independent patient. Paired operational comparisons should report effect sizes and exact p-values, with zero differences handled consistently in the signed-rank statistic and rank-biserial effect size. Repeated seeds and partitions are to be reported transparently, without declaring a method valid merely because a descriptive interval happens to contain the nominal level.

4.5. Reproducibility and Compute

The reference package requires Python 3.10 or later, with optional PyTorch and LightGBM dependencies. Threshold grids are fixed before risk calibration, and the deployment-support alarm bound includes both refractory and escalation settings. The conservative default in the generic budget driver allows any one-step positive exposure, while a larger independently declared

minimum is an explicit eligibility restriction. Version-two score caches reject obsolete unversioned scores, so earlier noncausal model outputs must be regenerated; exact clinical training versions, partitions, runtime and device measurements remain to be recorded when those experiments are executed.

References

[1] R. Zhang, F. Long, Z. Zhao, J. Wu, R. Tan, W. Xu, L. Li, Y. Long, H. Qu, Machine learning predicts sepsis deterioration trajectories, npj Digital Medicine 9 (1) (2026). doi:10.1038/s41746-026-02565-x.
URL https://doi.org/10.1038/s41746-026-02565-x

[2] A. Taparugssanagorn, M. Särestöniemi, M. Hämäläinen, J. Iinatti, Lightweight, explainable, and efficiency-aware deep learning for early sepsis prediction from structured electronic health record time-series, in: Communications in Computer and Information Science, Springer Nature Switzerland, 2026, pp. 225–244. doi:10.1007/978-3-032-28819-6_15.
URL https://doi.org/10.1007/978-3-032-28819-6_15

[3] L. Yong, L. Zhenzhou, Deep learning-based prediction of in-hospital mortality for sepsis, Scientific Reports 14 (1) (2024). doi:10.1038/s41598-023-49890-9.
URL https://doi.org/10.1038/s41598-023-49890-9

[4] K. Wang, H. Yan, D. Chen, X. Tan, H. Chen, X. Lu, H. Duan, S. Nan, Deep learning for early prediction of adverse events in intensive care units using electronic health records: A methodological review, Risk Management and Healthcare Policy 19 (2026) 1–22. doi:10.2147/rmhp.s629368.
URL https://doi.org/10.2147/rmhp.s629368

[5] C. A. Hernandez-Diaz, B. Vazquez, G. Fuentes-Pineda, Deep learning for multivariate time series analysis in electronic health records: a scoping review, Evolutionary Intelligence 19 (5) (2026). doi:10.1007/s12065-026-01246-8.
URL https://doi.org/10.1007/s12065-026-01246-8

[6] L. Swinckels, F. C. Bennis, K. A. Ziesemer, J. F. M. Scheerman, H. Bijwaard, A. de Keijzer, J. J. Bruers, The use of deep learning and machine

learning on longitudinal electronic health records for the early detection and prevention of diseases: Scoping review, Journal of Medical Internet Research 26 (2024) e48320. doi:10.2196/48320.
URL https://doi.org/10.2196/48320

[7] J. Kim, S. Park, Retamamba: Reliability-aware temporal aggregation with mamba for irregular clinical time series prediction, arXiv:2605.16380 (2026). arXiv:2605.16380, doi:10.48550/arxiv.2605.16380.
URL https://doi.org/10.48550/arxiv.2605.16380

[8] T. Shen, Z. Li, Y. Li, X. Qiu, X. Wu, B. Yang, J. Hu, Enhancing irregular time series forecasting with continuous-time modeling framework, arXiv:2607.28035 (2026). arXiv:2607.28035, doi:10.48550/arxiv.2607.28035.
URL https://doi.org/10.48550/arxiv.2607.28035

[9] V. K. Yalavarthi, K. Madhusudhanan, R. Scholz, N. Ahmed, J. Burchert, S. Jawed, S. Born, L. Schmidt-Thieme, Grafiti: Graphs for forecasting irregularly sampled time series, Proceedings of the AAAI Conference on Artificial Intelligence 38 (15) (2024) 16255–16263. doi:10.1609/aaai.v38i15.29560.
URL https://doi.org/10.1609/aaai.v38i15.29560

[10] Y. Mun, J. H. Ahn, C. Y. Lee, B. Ahn, J. W. Huh, K. S. Chung, Made: leveraging informative missingness and temporal evaluation for electronic health record-based deterioration prediction in general wards, Journal of the American Medical Informatics Association (2026). doi:10.1093/jamia/ocag124.
URL https://doi.org/10.1093/jamia/ocag124

[11] O. O. Komolafe, Z. Mei, D. M. Zarate, G. W. Spangenberg, Early prediction of sepsis: Feature-aligned transfer learning, arXiv:2505.02889 (2025). arXiv:2505.02889, doi:10.48550/arxiv.2505.02889.
URL https://doi.org/10.48550/arxiv.2505.02889

[12] C. Yin, S. Fu, B. Yao, T.-H. Pham, W. Cao, D. Wang, J. Caterino, P. Zhang, Sepsiscalc: Integrating clinical calculators into early sepsis prediction via dynamic temporal graph construction, in: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and

Data Mining V.1, ACM, 2025, pp. 2779–2790. doi:10.1145/3690624.3709402.
URL https://doi.org/10.1145/3690624.3709402

[13] L. J. L. López, F. E. Shamout, T. G. J. Rudner, An empirical analysis of calibration and selective prediction in multimodal clinical condition classification, arXiv:2603.02719 (2026). arXiv:2603.02719, doi:10.48550/arxiv.2603.02719.
URL https://doi.org/10.48550/arxiv.2603.02719

[14] F. Cenacchi, L. Cao, R. Yang, Beyond output-space calibration: Spectral evidence bundling for selective reliability estimation in time-series classification, arXiv:2607.18279 (2026). arXiv:2607.18279, doi:10.48550/arxiv.2607.18279.
URL https://doi.org/10.48550/arxiv.2607.18279

[15] J. Strickland, M. Ghisoni, H. Marshall, T. Whitehead, B. Nenchev, B. Pellegrini, C. Phillips, K. Tassenberg, S. Davey, S. Dorman, J. Sol, D. Ferguson, G. Conduit, Degrees of uncertainty: conformal deep learning for non-invasive core body temperature prediction in extreme environments, Communications Engineering 4 (1) (2025). doi:10.1038/s44172-025-00548-6.
URL https://doi.org/10.1038/s44172-025-00548-6

[16] M. H. Mehdizavareh, G. Santangelo, G. Nicora, S. L. Cichosz, A. Dagliati, A. Khan, R. Bellazzi, Informative missingness to generate irregular clinical time series, arXiv:2606.17106 (2026). arXiv:2606.17106, doi:10.48550/arxiv.2606.17106.
URL https://doi.org/10.48550/arxiv.2606.17106

[17] C. Xu, R. Yang, X. Xiang, Y. Wu, Y. Huang, H. Lei, J. Yang, Bridging the data scarcity in venous thromboembolism detection: A deep learning framework for large-scale irregular clinical time series, in: Proceedings of the Thirty-Fifth International Joint Conference on Artificial Intelligence, International Joint Conferences on Artificial Intelligence Organization, 2026, pp. 6591–6599. doi:10.24963/ijcai.2026/733.
URL https://doi.org/10.24963/ijcai.2026/733

[18] C. Sun, M. Song, D. Cai, B. Zhang, H. Li, S. Hong, A review of deep learning methods for irregularly sampled medical time series data,

Health Data Science 6 (2026). doi:10.34133/hds.0456.
URL https://doi.org/10.34133/hds.0456

[19] W. Nie, Z. Qu, W. Wang, C. Li, K. Lu, B. Zhou, H. Yu, Clinically interpretable sepsis early warning via llm-guided simulation of temporal physiological dynamics, arXiv:2604.20924 (2026). arXiv:2604.20924, doi:10.48550/arxiv.2604.20924.
URL https://doi.org/10.48550/arxiv.2604.20924

[20] R. Rong, Z. Gu, H. Lai, T. L. Nelson, T. Keller, C. Walker, K. W. Jin, C. Chen, A. M. Navar, F. Velasco, E. D. Peterson, G. Xiao, D. M. Yang, Y. Xie, A deep learning model for clinical outcome prediction using longitudinal inpatient electronic health records, JAMIA Open 8 (2) (2025). doi:10.1093/jamiaopen/ooaf026.
URL https://doi.org/10.1093/jamiaopen/ooaf026

[21] F. Mohammad, L. Arunachalam, S. Sadhu, B. Aasman, S. Garg, A. Ahmed, S. Colman, M. Arunachalam, S. Kulkarni, P. Mirhaji, Early prediction of onset of sepsis in clinical setting, arXiv:2402.03486 (2024). arXiv:2402.03486, doi:10.48550/arxiv.2402.03486.
URL https://doi.org/10.48550/arxiv.2402.03486

[22] M. R. Fentazi, M. Ameur, A. Ksentini, Training on the future: A delay-aware audit of test-time adaptation for time-series forecasting, arXiv:2610.12232 (2026). arXiv:2610.12232, doi:10.48550/arxiv.2610.12232.
URL https://doi.org/10.48550/arxiv.2610.12232

[23] K. Li, Y. Zhan, H. Jin, C. Qi, X. Lin, B. Yu, Cross-sample augmented test-time adaptation for personalized intraoperative hypotension prediction, Proceedings of the AAAI Conference on Artificial Intelligence 40 (27) (2026) 22958–22966. doi:10.1609/aaai.v40i27.39460.
URL https://doi.org/10.1609/aaai.v40i27.39460

[24] K. G. M. Moons, J. A. A. Damen, T. Kaul, L. Hooft, C. A. Navarro, P. Dhiman, A. L. Beam, B. V. Calster, L. A. Celi, S. Denaxas, A. K. Denniston, M. Ghassemi, G. Heinze, A. P. Kengne, L. Maier-Hein, X. Liu, P. Logullo, M. D. McCradden, N. Liu, L. Oakden-Rayner, K. Singh, D. S. Ting, L. Wynants, B. Yang, J. B. Reitsma, R. D. Riley, G. S. Collins, M. van Smeden, Probast+ai: an updated quality, risk

of bias, and applicability assessment tool for prediction models using regression or artificial intelligence methods, BMJ 388 (2025) e082505. doi:10.1136/bmj-2024-082505.
URL https://doi.org/10.1136/bmj-2024-082505